Simple Unit Root Tests

Unit Root Tests

- Stationarity is fundamentally important for time series analysis. If a time series is not stationary, we must transform the nonstationary series to a stationary series before performing the regression analysis. Otherwise, the regression result is not reliable.

- Usually, taking the (first) difference or natural log on the time variable is one of the possible ways to stabilize the series, i.e., $\Delta y_t = y_t - y_{t-1}$ or $\Delta \ln(y_t)$

- Graphically, if the ACF value of the time series y_t does not die out quick enough after many lags, then the series is likely a nonstationary series.

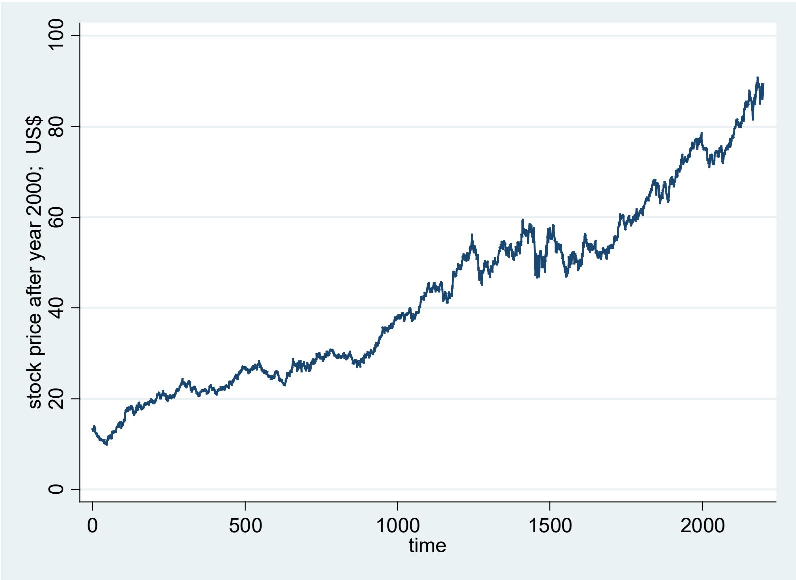
- On the other hand, for stationary variables, their ACF values decline exponentially when the time distance h increases.

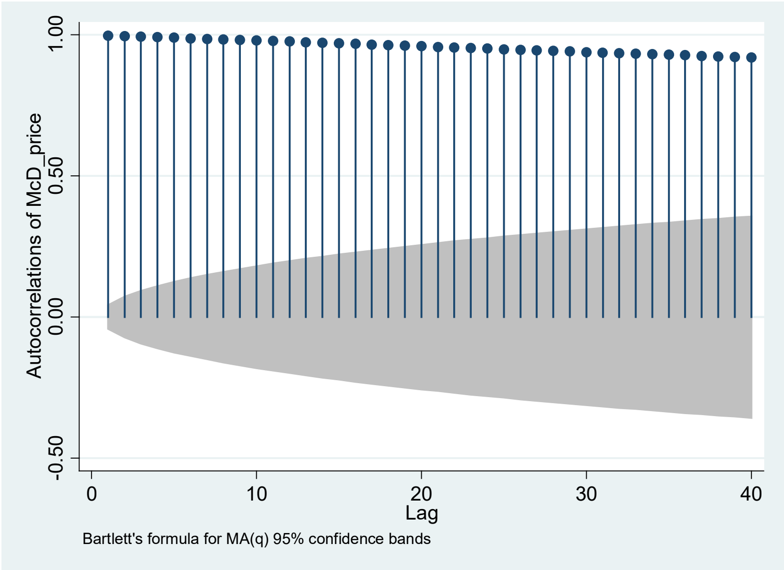
Example: We analyze the McDonald's closing stock price.

Stata> tsset time

Stata> tsline McD_price

Stata> ac McD_price





Both diagrams hint that the stock price series is nonstationary. Graphically, for nonstationary variables, the ACF does not die out easily. (very persistent!).

- What if we use $AR(1)$ to model the McDonald's stock price?

```
. reg  McD_price l1.McD_price

      Source |       SS       df       MS              Number of obs =    2199
-------------+------------------------------           F(  1,  2197) =       .
       Model |  876931.814     1  876931.814           Prob > F      =  0.0000
    Residual |  872.064398  2197  .396934182           R-squared     =  0.9990
-------------+------------------------------           Adj R-squared =  0.9990
       Total |  877803.879  2198  399.364822           Root MSE      =  .63003

------------------------------------------------------------------------------
   McD_price |      Coef.   Std. Err.      t    P>|t|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
   McD_price |
         L1. |   1.000243   .0006729  1486.36   0.000     .9989235    1.001563
             |
       _cons |   .0242001   .0317506     0.76   0.446    -.0380642    .0864644
------------------------------------------------------------------------------
```

- The statistical report shows that R-squared is very high, and (b) the estimated coefficient is bigger than 1. However, this report is not reliable.

- The regression model is wrong, and it will eventually "explode".

- When the coefficient is 1 or larger than 1, the variable is nonstationary, and the regression result is not reliable. Usually, its R-squared value is very high, but it is misleading. This type of regressions is called spurious regression.

- How do we statistically test if the time series data y_t is (non)stationary?
  - Suppose we have an $AR(1)$ model, $y_t = \rho y_{t-1} + u_t$
  - If $\rho = 1$, a unit root exits, i.e., the series is not stationary.

- Therefore, we can test if ρ is significantly different from 1.

- Alternatively, we can run the regression: $y_t - y_{t-1} = (\rho - 1)\, y_{t-1} + u_t$ and test if $(\rho - 1)$ is zero.

Dickey Fuller Test

$y_t - y_{t-1} = (\rho - 1)\, y_{t-1} + u_t$

$\Rightarrow \Delta y_t = \delta y_{t-1} + u_t$

H_0: $\delta = 0$ (unit root is present) versus H_1: no unit root/ series is stationary

- There are three main versions of the Dickey Fuller test, DF.

Case 1: $\Delta y_t = \delta y_{t-1} + u_t$,

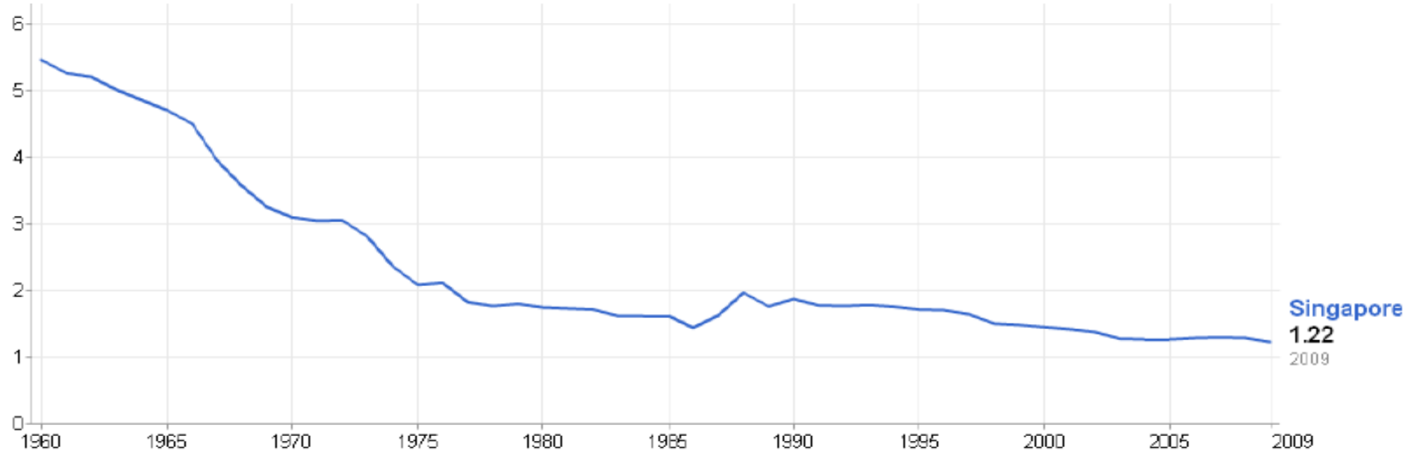
Case 2: $\Delta y_t = \beta_0 + \delta y_{t-1} + u_t$

Case 3: $\Delta y_t = \beta_0 + \beta_1 t + \delta y_{t-1} + u_t$

- Whether to include the intercept and/or time trend is an empirical question.

- This DF test is very restricted, because it applies only on the $AR(1)$.

Example: Fertility Rate



Example: Singapore Inflation Rate (1981 Oct to 2015 Dec)

Stata> dfuller inf, regress /* case (2) */

```
Dickey-Fuller test for unit root                   Number of obs   =       410

                               ---------- Interpolated Dickey-Fuller ---------
                  Test         1% Critical       5% Critical      10% Critical
               Statistic           Value             Value             Value
------------------------------------------------------------------------------
 Z(t)             -4.009            -3.447            -2.874            -2.570
------------------------------------------------------------------------------
MacKinnon approximate p-value for Z(t) = 0.0014
```

Stata> dfuller inf, regress trend /* case (3) */

/* inf (inflation) is the variable name; I include time trend and 1 lag in the unit root test */

```
Dickey-Fuller test for unit root                   Number of obs   =       410

                               ---------- Interpolated Dickey-Fuller ---------
                  Test         1% Critical       5% Critical      10% Critical
               Statistic           Value             Value             Value
------------------------------------------------------------------------------
 Z(t)             -4.029            -3.984            -3.424            -3.130
------------------------------------------------------------------------------
MacKinnon approximate p-value for Z(t) = 0.0080
```

Stata> dfuller inf, regress noconstant /* case (1) */

We reject the null. The series can be assumed as a stationary series.

Augmented Dickey Fuller Test

- The DF test applies only on the $AR(1)$, and the $AR(1)$ might not capture all serial correlation in y_t in which case the $AR(p)$ is more appropriate.

- The Augmented Dickey Fuller test (ADF) is a more general unit root test compared as it can be used to test for the $AR(p)$ process.

The unit test model is as follow.

$$\Delta y_t = \delta y_{t-1} + \gamma_1 \Delta y_{t-1} + \cdots + \gamma_p \Delta y_{t-p} + u_t$$

H_0: $\delta = 0$ (unit root is present) versus H_1: H_0 is wrong

- Similarly, we can include an intercept and/or a time trend into the model.

Case (1): $\Delta y_t = \delta y_{t-1} + \gamma_1 \Delta y_{t-1} + \cdots + \gamma_p \Delta y_{t-p} + u_t$

Case (2): $\Delta y_t = \beta_0 + \delta y_{t-1} + \gamma_1 \Delta y_{t-1} + \cdots + \gamma_p \Delta y_{t-p} + u_t$

Case (3): $\Delta y_t = \beta_0 + \beta_1 t + \delta y_{t-1} + \gamma_1 \Delta y_{t-1} + \cdots + \gamma_p \Delta y_{t-p} + u_t$

- We can graph the series y_t, and decide which model (case 1, 2, or 3) to use. But, how many lags of Δy_t should we use the test model?

- Schwert (1989) suggested that one can set the number of lags not bigger than p_{max}, where $p_{max} = \left(12 \times (\frac{n}{100})^{\frac{1}{4}}\right)$

Example:
If there are 410 monthly inflation observations, then we can set the number of lags in the ADF test at most up to $p_{max} = 12 \times (\frac{410}{100})^{\frac{1}{4}} \approx 17.$

If the estimated slope of Δy_{t-17} is not significant, then we perform the unit root test again by using 16 lags.

Keep reducing the # of lags until the estimated slope of Δy_t is significant.

At first, we include 17 lags. Stata > dfuller inf, regress lags(17)

The inflation rate variable doesn't contain unit root. It is a stationary series.

```
Augmented Dickey-Fuller test for unit root          Number of obs   =       393

                               ---------- Interpolated Dickey-Fuller ---------
                  Test         1% Critical       5% Critical      10% Critical
               Statistic           Value             Value             Value
------------------------------------------------------------------------------
 Z(t)             -3.502            -3.449            -2.874            -2.570
------------------------------------------------------------------------------
MacKinnon approximate p-value for Z(t) = 0.0079
```

D.inf	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
inf						
L1.	-.0569843	.0162738	-3.50	0.001	-.0889838	-.0249847
LD.	-.0089345	.0514374	-0.17	0.862	-.1100773	.0922083
L2D.	.122671	.051432	2.39	0.018	.0215389	.2238031
L3D.	.1750556	.0516876	3.39	0.001	.0734208	.2766905
L4D.	.0454051	.0524244	0.87	0.387	-.0576785	.1484886
L5D.	.1468305	.0523328	2.81	0.005	.0439272	.2497338
L6D.	.0666337	.0453375	1.47	0.142	-.0225147	.1557822
L7D.	.0165462	.0454432	0.36	0.716	-.07281	.1059025
L8D.	.0298502	.0453246	0.66	0.511	-.0592728	.1189731
L9D.	.1437163	.044735	3.21	0.001	.0557526	.23168
L10D.	.0332401	.0452344	0.73	0.463	-.0557056	.1221858
L11D.	.0493367	.0448474	1.10	0.272	-.038848	.1375214
L12D.	-.4441752	.043917	-10.11	0.000	-.5305305	-.35782
L13D.	-.042234	.049081	-0.86	0.390	-.1387432	.0542752
L14D.	.0665632	.0487572	1.37	0.173	-.0293094	.1624358
L15D.	.0981115	.0483171	2.03	0.043	.0031043	.1931186
L16D.	.0340055	.0480826	0.71	0.480	-.0605405	.1285516
L17D.	.0322537	.0480884	0.67	0.503	-.062304	.1268113
_cons	.0949356	.0369476	2.57	0.011	.0222846	.1675866

Appendix: Other unit root tests

- The Phillips-Perron unit root test (PP) is very popular in the analysis of financial time series. Unlike the DF and ADF tests, the PP test corrects for any unspecified type of serial correlation and heteroskedasticity in the errors, u_t.

- Thus, in practice, the PP test is better than the ADF test, because the test is more robust to the violations of classical linear assumptions. In addition, we do not need to specify the number of lags.

Example: Inflation rate

Stata>pperron inf

```
. pperron inf

Phillips-Perron test for unit root                 Number of obs   =       410
                                                   Newey-West lags =         5

                                   ---------- Interpolated Dickey-Fuller ---------
                  Test             1% Critical       5% Critical      10% Critical
               Statistic              Value             Value             Value
------------------------------------------------------------------------------
 Z(rho)         -29.100            -20.428           -14.000           -11.200
 Z(t)            -4.378             -3.447            -2.874            -2.570
------------------------------------------------------------------------------
MacKinnon approximate p-value for Z(t) = 0.0003
```

The p-value for the test statistic is smaller than the significance levels at 1%, 5% and 10%. We reject the null hypothesis.

- Kwiatkowski, Phillips, Schmidt and Shin (KPSS) developed a unit root test to complement to the traditional unit root tests, such as ADF and PP tests.

- For ADF, PP tests – $H_0: \delta = 0$ (unit root is present) versus $H_1: \delta < 0$

- For KPSS test -- H_0: there is not unit root versus H_1: there is a unit root. Statistically speaking, the ADF (or PP) test is to find evidence to reject unit root, and the KPSS test is to find evidence to reject no unit root.

Example: Stata> kpss inf

```
KPSS test for inf

Maxlag = 17 chosen by Schwert criterion
Autocovariances weighted by Bartlett kernel

Critical values for H0: inf is trend stationary

10%: 0.119   5% : 0.146   2.5%: 0.176   1% : 0.216

Lag order    Test statistic
    0            1.12
    1            .579
    2            .395
    3            .304
    4             .25
    5            .215
    6             .19
    7            .171
    8            .157
    9            .146
   10            .138
   11             .13
   12            .125
   13             .12
   14            .116
   15            .112
   16            .109
   17            .107
```